PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Cost Effectiveness of Treatment Optimization with Biomarkers for Immunotherapy in Solid Tumors: A Systematic Review Protocol
AUTHORS	Mucherino, Sara; Lorenzoni, Valentina; Orlando, Valentina; Triulzi, Isotta; Del Re, Marzia; Capuano, Annalisa; Danesi, Romano; Turchetti, Giuseppe; Menditto, Enrica

VERSION 1 – REVIEW

REVIEWER	Gonalez-Ericsson, Paula I Vanderbilt University Medical Center, Vanderbit Ingram Cancer Center
REVIEW RETURNED	22-Feb-2021

GENERAL COMMENTS	The analysis of the feasibility and cost-effectiveness of implementing therapies and their biomarkers into the real clinical world is of paramount importance. The objectives of this study are very relevant, and I commend the authors on embarking on such a task, however, some clarifications on the current protocol are needed. My main concerns are: 1. The authors must clarify what type of biomarkers will be included in the review. The abstract and lines 173 and 189 mention blood-based biomarkers, however, there is no mention of limitation to blood-base biomarkers in any other part of the manuscript and no specific limitation to blood-base biomarkers within the search terms, which will include tissue-based and clinical biomarkers. Along the same line, the introduction mentions a genetic-test as complementary/companion tests listed by the FDA, but complementary/companion tests are not only genetic. Is this specifically mentioned because the authors will only be focusing on genetic tests? Later in the introduction authors mention PD-L1, which is commonly evaluated by IHC on tissue samples. If only blood-based biomarkers will be searched, explain why these were chosen. 2. Table 2 needs some clarifications: a. Patient diagnosis: should include tumor type, location, and primary vs met status. b. Patient: “solid tumors treated with immunotherapy” is not clear. Do the authors mean Type of therapy? Should identify specific immunotherapy drug +/- combination therapy. c. Intervention: For each biomarker, type of test and definition of biomarker positivity should be included. d. Comparator: any other strategies. Would this be the use of the therapy without a biomarker? Which other strategies are expected? 3. Authors need to specify how cost-effectiveness, cost-utility, and net-monetary benefit will be evaluated. Will the cost-effectiveness of the therapy also be evaluated and included as part of the

	biomarker cost-effectiveness? If different sources use different cost-effectiveness quantifications, how will these be compared? What type of comparisons are planned? Use of biomarker vs no biomarker? Will the authors be comparing the cost-effectiveness of a specific biomarker across tumor types and of different biomarkers with-in a tumor type? 4. I would suggest separating clinical benefit (life-years gained and quality of life) from the economic-related outcomes. 5. Will the authors include an evaluation of early neoadjuvant and late-stage cancer therapy? If so, clinical benefit in the neoadjuvant setting could be measured in different parameters. How will this be addressed? 6. I would recommend extending the finish date for database queries (Line 111). Cancer immunotherapy is a growing area of interest that is continuously evolving, relevant information would be missing from the review if the search is capped at June 2020. 7. I would consider the availability of the full text published in English a language restriction (Line 138). In fact, one of the references of the current manuscript (10) does not fulfill this requirement. Some minor concerns: 8. Wording needs editing. Verb tenses must be consistent. 9. Line 70: This sentence is not clear. Do the authors mean that out of all the complementary/companion diagnostic tests approved by the FDA (48), only 29% impact the physician's choice of whether to prescribe a certain drug. It would be interesting for the authors to include the numbers that pertain specifically to immunotherapy.

REVIEWER	Huxley, Nicola Monash University, Centre for Health Economics, Monash Business School
REVIEW RETURNED	01-Apr-2021

GENERAL COMMENTS	Thank you for this protocol to investigate the cost-effectiveness of biomarkers in the treatment of cancer. I think with a few changes this systematic review could provide an important addition to the literature. My main concerns are a lack of detail within the protocol, specifically with regards the justification of the paper, what the comparators are and how the studies will be synthesised. Also it would be good to update the methods with reference to the updated 2020 PRISMA guidelines. I detail some specific comments below. Abstract, p2: Please be clear whether comparators are targeted therapies (using clinical indicators) or non-targeted therapies- the comparator is as likely to drive cost-effectiveness as the intervention. Cost-effectiveness and cost-utility are study types, not outcomes. For this study you may prefer to use the PICOS layout with outcomes including costs, LYs, QALYs, ICERs, net health benefit, net monetary benefit, CEAC, CEAF; and study types including cost-effectiveness, cost-utility, economic evaluation. You might also want to consider using an economic evaluation specific checklist to assess the quality of the included studies, such as CHEERS Introduction, p4, line 70: "29% of cases" I'm not sure what these cases are that are referred to here. Please clarify.

	Introduction, p4, line 81: it would be good to be a bit clearer as to why these specific biomarkers over others. Introduction, p5, line 91: “maximise clinical benefit” vs what comparator? There needs to be a discussion of what the targeted biomarkers replace (non-personalised therapies or individualised treatment using clinical indicators) Introduction, p5, line 97-99: there might not be any published synthesis of these studies, but there are a lot of studies for each indication, so how will the synthesis add to the literature. In particular, what is the justification for synthesising individualised treatments? If they are so personalised, can they be synthesised? Methods, Search Strategy, p6, line 116: searching only title/abstract will make your search more specific but less sensitive. Please add a justification for restricting to title/abstract searching. Methods, Search Strategy, p6, line 117: please provide some information on how the list of interventions were identified. Also Nivolumab is mentioned twice. Methods, Search Strategy, p6, line 120: It would be good to know how search strategy was developed, e.g. in conjunction with clinicians/information scientists, adapting a previous search, etc. Methods, Search Strategy, p6: Will you also be searching HTA reports for various countries via websites such as NICE? It would also be good to justify why you are restricting your search to PubMed and Embase, when there are other databases. Using only two may make your search less sensitive- how confident are you that all studies will be captured? Methods, Eligibility criteria, p6, lines 123-130: Check spelling of “Eligibility”. As mention under my Abstract comments, it would be good to be clearer about the comparators, and use a PICOS structure, where cost-effectiveness and cost-utility are study designs not outcomes (but with the caveat that adding a study design criteria makes search less sensitive) Methods, Eligibility criteria, p7, line 132: I couldn’t find mention of the date restriction in the search strategy appendix. Perhaps you could add this in. Also do you have plans to update the search if there are delays to the review/prior to publication? Methods, Eligibility criteria, p7, lines 135-137: These are exclusions by study type, which reinforces that you are using more of a PICOS structure. You might want to consider a summary table of inclusion and exclusion criteria for clarity. Methods, Eligibility criteria, p7 lines 138-140: if only full texts in English are included, this is a language restriction. It is misleading to begin this sentence saying there is no language restriction Methods, Selection and data process, p7, line 146: Please describe how the 4 researchers will divide up screening. Will all 4 review everything, or will there be double screening of sections of the search? What will be your exclusion criteria at the title/abstract stage compared to full text? Methods, Selection and data process, p7, line 149-150: backward reference searching is described here, but will you also do forward citation searching (seeing what studies cite that paper)? Will there also be searching of grey literature (like the unpublished HTA reports described above)? Methods and Analysis, Selection and data process, p8, lines 154-158: You may also wish to consider the use of an economic specific checklist like CHEERS. The EQUATOR network https://www.equator-network.org/ is good for identifying relevant checklists for different paper types that may help you evaluate published studies as much as write your own

	Methods and Analysis: I would recommend producing for the appendix an example of your data extraction tables and any summary tables that will compare study characteristics, results and study quality. You can then reference these in the methods section Ethics and Dissemination: I think this section reads more like part of the methods, with some explanation of how you will present/interpret the results and therefore the heading does not seem appropriate. You may wish to write a separate section focussed on ethics and dissemination, and title this section as part of the methods (evidence synthesis?) Ethics and Dissemination, p8, line 168-169: better cost-effectiveness will depend entirely on how the study judges cost-effectiveness, e.g., willingness to pay thresholds, what the comparator was. It would be good to describe how you will synthesise different measures of cost-effectiveness particularly across different healthcare systems. Ethics and Dissemination, p8, line 174: “diffusion of real use” How do you intend to assess this? E.g., will you be comparing trial based evaluations to registry based evaluations to evaluations modelled using estimates from published literature and comparing the use to guidelines produced in the country of the study? Ethics and Dissemination, p9, line 183: This is the first real mention of how you might synthesise the data. Please expand upon this with details of what study characteristics, outcomes you might synthesise and how this might be achieved if the studies are too heterogeneous to warrant a full meta-analysis (could you split into anticipated subgroups for analysis, or would you narratively synthesise everything?) References, p11-14: add PRISMA 2020 and refer to it in the methods study development. Also you should address in the text how your study differs from Oosterhoff et al. 2016 Tables, Table 1, p15: It would be good if the search strategy and the PICO criteria lined up, currently the relationship is not clear. You might also want to add the date restrictions, and expand some of the search terms. For example, quality of life might be capture with synonymous terms/measures etc., such as health utility, quality adjusted life years, QALYs, utility, patient reported outcomes, EQ-5D, SF-36. You are also missing several stated outcomes, such as life years (or LYs, life years gained, LYG), net monetary benefit, net health benefit, ICERs. Cost-benefit analysis, cost-effectiveness, cost-utility and economic evaluation are all study types, not outcomes and while it may be justified to include them to make your search more specific, it will make your search less sensitive. I think you should also consider what you are interested in as comparators and include them in your search. Reporting checklist for protocol of a systematic review: Potential amendments are not explicitly stated in the text Objectives: comparators and outcomes are not clear (comparators not specific, study designs incorrectly labelled as outcomes) Data items: no pre-planned data assumptions or simplifications are stated, nor specific variables of interest (e.g. ICER, willingness to pay, costs) Outcomes and prioritization: As above, it is not clear which outcomes are to be prioritised. Cost/QALY gained? Net health benefit? Costs? Data synthesis 15a-15d: There is very little description of data synthesis in the text, no description of summary measures, or

	potential subgroups or alternatives to quantitative synthesis if that becomes inappropriate Despite my requests for clarification, I do believe this systematic review could become an informative piece of work. However, at the moment I think more needs to be done in defining the methodology and how the results will be synthesised. I also have concerns that the current search strategies are not suitably justified and may miss some publications. It would be good to look at previous economic search strategies (either in publications or unpublished HTA reports) or involve an information scientist to help identify additional terms for the searches if not done already.

VERSION 1 – AUTHOR RESPONSE

RESPONSES TO REVIEWER 1 COMMENTS

Dr. Paula I Gonalez-Ericsson, Vanderbilt University Medical Center

- **The analysis of the feasibility and cost-effectiveness of implementing therapies and their biomarkers into the real clinical world is of paramount importance. The objectives of this study are very relevant, and I commend the authors on embarking on such a task, however, some clarifications on the current protocol are needed. My main concerns are:**

Answer: Dear Reviewer, on behalf of all the authors, we really appreciate your interest in the study.

- **The authors must clarify what type of biomarkers will be included in the review. The abstract and lines 173 and 189 mention blood-based biomarkers, however, there is no mention of limitation to blood-base biomarkers in any other part of the manuscript and no specific limitation to blood-base biomarkers within the search terms, which will include tissue-based and clinical biomarkers. Along the same line, the introduction mentions a genetic-test as complementary/companion tests listed by the FDA, but complementary/companion tests are not only genetic. Is this specifically mentioned because the authors will only be focusing on genetic tests? Later in the introduction authors mention PD-L1, which is commonly evaluated by IHC on tissue samples. If only blood-based biomarkers will be searched, explain why these were chosen.**

Answer: Thanks for the warning. We agree with your observation. Our intention is to focus on the use of biomarkers in patients with solid tumors treated with immune checkpoint inhibitors (ICIs) already approved, such as nivolumab, pembrolizumab, ipilimumab, atezolizumab, durvalumab, avelumab, cemiplimab. In light of this, biomarkers we will expect from the analysis are PD-1, PDL-1, CTLA-4. We specified this in the whole study protocol.

- **Table 2 needs some clarifications:**

a. Patient diagnosis: should include tumor type, location, and primary vs met status.

b. Patient: "solid tumors treated with immunotherapy" is not clear. Do the authors mean Type of therapy? Should identify specific immunotherapy drug +/- combination therapy.

c. Intervention: For each biomarker, type of test and definition of biomarker positivity should be included.

d. Comparator: any other strategies. Would this be the use of the therapy without a biomarker? Which other strategies are expected?

Answer: Thanks for the suggestions, we detailed and implemented the PICOS criteria, as you can see in Table 2 (now Table 3).

- **Authors need to specify how cost-effectiveness, cost-utility, and net-monetary benefit will be evaluated. Will the cost-effectiveness of the therapy also be evaluated and included as part of the biomarker cost-effectiveness? If different sources use different cost-effectiveness quantifications, how will these be compared? What type of comparisons are planned? Use of biomarker vs no biomarker? Will the authors be comparing the cost-effectiveness of a specific biomarker across tumor types and of different biomarkers within a tumor type?**

Answer: We provided a more detailed specification of what we expect from outcomes (Table 3). We then clarified that health economic outcomes (incremental cost-effectiveness ratio, net health benefit, net monetary benefit, life-years and quality adjusted life years) will be evaluated between immune checkpoint inhibitors therapy with and without previous biomarker's test and other conventional targeted and non targeted-therapy

- **I would suggest separating clinical benefit (life-years gained and quality of life) from the economic-related outcomes.**

Answer: Thanks, we agree with your suggestion so we separated the two diffent types of outcomes as economic and clinical, see Table 3.

- **Will the authors include an evaluation of early neoadjuvant and late-stage cancer therapy? If so, clinical benefit in the neoadjuvant setting could be measured in different parameters. How will this be addressed?**

Answer: We do not plan to differentiate the tumor stage. We mainly intend to observe differences in economic and clinical benefit of biomarker use before initiation of therapy, not discriminating by disease stage; we will consider stratifing results for disease stage depending on the references retrieved and the availability of information,

- **I would recommend extending the finish date for database queries (Line 111). Cancer immunotherapy is a growing area of interest that is continuously evolving, relevant information would be missing from the review if the search is capped at June 2020.**

Answer: We completely understand and agree with your opinion. June 2020 was set as final date as the first submission of the protocol was done in December 2020. Therefore, we have extended the end date of the analysis to June 2021. This was also changed in the study protocol.

- **I would consider the availability of the full text published in English a language restriction (Line 138). In fact, one of the references of the current manuscript (10) does not fulfill this requirement.**

Answer: Thanks for the suggestion. We rephrased as follow: “Moreover, language restriction will be applied to the research, as fundamental to the eligibility of the study will be the availability of the papers’ full text published in English.”

- **Wording needs editing. Verb tenses must be consistent.**

Answer: We reviewed and reread the entire manuscript making all appropriate corrections.

- **Line 70: This sentence is not clear. Do the authors mean that out of all the complementary/companion diagnostic tests approved by the FDA (48), only 29% impact the physician’s choice of whether to prescribe a certain drug. It would be interesting for the authors to include the numbers that pertain specifically to immunotherapy.**

Answer: What we meant was that among the 166 biomarker-drug combinations reported by the FDA, in only 29% (48 combinations) of cases, results obtained from biomarker test have an impact on the physician's choice to prescribe or not prescribe a particular drug. We rephrased the sentence in the introduction section.

RESPONSES TO REVIEWER 2 COMMENTS

Dr. Nicola Huxley, Monash University

- **Thank you for this protocol to investigate the cost-effectiveness of biomarkers in the treatment of cancer. I think with a few changes this systematic review could provide an important addition to the literature.**

Answer: Dear Reviewer, we sincerely appreciate your interest in this research.

- **My main concerns are a lack of detail within the protocol, specifically with regards the justification of the paper, what the comparators are and how the studies will be synthesised. Also it would be good to update the methods with reference to the updated 2020 PRISMA guidelines.**

Answer: Dear Reviewer, please, consider the resubmitted paper fully revised. We have clarified both in the text and in Table 1 which comparators are in our analysis by changing the model followed to conduct the review, no longer PICO but PICOS model. We have specified that we will use the updated PRISIMA 2020 statement guidelines in conducting the review. In addition, we revised the search strategy by implementing it based on the PICOS model. Full details can be found in the manuscript and responses below.

- **Abstract, p2: Please be clear whether comparators are targeted therapies (using clinical indicators) or non-targeted therapies- the comparator is as likely to drive cost-effectiveness as the intervention.**

Answer: We have specified more clearly the comparator of our analysis. The comparator will be any other targeted or non-targeted therapy versus the intervention which instead will be each test of the immune checkpoint predictive biomarkers, such as PD1, PDL-1 CTLA4, IL-6. We specified this in the abstract, in the text, and in Table 3.

- **Cost-effectiveness and cost-utility are study types, not outcomes. For this study you may prefer to use the PICOS layout with outcomes including costs, LYs, QALYs, ICERs, net health benefit, net monetary benefit, CEAC, CEAF; and study types including cost-effectiveness, cost-utility, economic evaluation.**

Answer: We really agree with your observation. For this reason, we have rebuilt our PICO model by changing it to PICOS, as showed in Table 3 and specified in the text. We defined the PICOS Model as follow:

Patient (P): Patients with solid tumors treated with immune checkpoint inhibitors (monotherapy or combination therapy): nivolumab, pembrolizumab, ipilimumab, atezolizumab, durvalumab, avelumab, cemiplimab; Intervention (I): Test of the immune checkpoint predictive biomarkers, such as PD1, PDL-1 CTLA4, IL-6. Comparator (C): Any other targeted or non-targeted therapy; Outcomes (O): Economic outcomes (Cost-effectiveness ratio (ICERs), net monetary benefit, LYs, QALYs) will be evaluated between immune checkpoint inhibitors therapy with and without previous biomarker's test. Study design (S): Economic evaluations reporting cost-effectiveness analysis, cost-utility analysis, net-monetary benefit.

- **You might also want to consider using an economic evaluation specific checklist to assess the quality of the included studies, such as CHEERS**

Answer: Thanks for the useful suggestion. We will use the CHEERS checklist to assess the quality of the economic evaluations that will be included in the analysis. We have also specified this in the methods section, in the Elegibility criteria sub-section.

- **Introduction, p4, line 70: "29% of cases" I'm not sure what these cases are that are referred to here. Please clarify.**

Answer: We clarified and rephrased the sentence in the introduction section as follow: "among the 166 biomarker-drug combinations reported by the FDA, in only 29% (48 combinations) of cases, results obtained from biomarker test have an impact on the physician's choice to prescribe or not prescribe a particular drug".

- **Introduction, p4, line 81: it would be good to be a bit clearer as to why these specific biomarkers over others.**

Answer: Dear Reviewer, thanks for the suggestion. We specified in the introduction section section why we decided to focus our analysis specifically on predicitive biomarkers in the frameswork of the treatment with immune-checkpoint inhibitors (ICIs), which will be firstly approved by the FDA and than by the Italian Medicine Agency, for the treatment of several solid tumors. So, our analysis will aim to investigate how these biomarker tests will be useful in the immunotherapy in terms of cost and clinical benefit. Reason why the biomarkers expected in the analysis are PD-1, PLD-1, CTLA-4 wich are crucial for the immune checkpoint.

- **Introduction, p5, line 91: "maximise clinical benefit" vs what comparator? There needs to be a discussion of what the targeted biomarkers replace (non-personalised therapies or individualised treatment using clinical indicators)**

Answer: Comparators meant in this sentence were not-targeted therapy, reason why we rephared the sentence in the introduction section as follow: A targeted approach to treatment using predictive biomarkers has the potential not only to maximize clinical benefit in respect to the not-target therapy, but also to improve cost-effectiveness and reduce the economic burden of the disease".

- **Introduction, p5, line 97-99: there might not be any published synthesis of these studies, but there are a lot of studies for each indication, so how will the synthesis add to the literature. In particular, what is the justification for synthesising individualised treatments? If they are so personalised, can they be synthesised?**

Answer: Thanks for your comment. It is likely that our research will produce many heterogeneous studies that will make it difficult to synthesize results, but will allow a description of the use of predictive biomarkers in immunotherapy in clinical and economic terms compared to non-personalized therapies. We have for this reason eliminated the term 'synthesize'.

- **Methods, Search Strategy, p6, line 116: searching only title/abstract will make your search more specific but less sensitive. Please add a justification for restricting to title/abstract searching.**

Answer: We will not search only title and abstract but the search startegy will combine both Mesh Terms (via PubMed) or Emtree terms (via Embase) and title and abstract. The best match will be included in the final search strategy.

- **Methods, Search Strategy, p6, line 117: please provide some information on how the list of interventions were identified. Also Nivolumab is mentioned twice.**

Answer: Dear Reviewer, we have corrected the mistake and implemented the terms of the search strategy by inserting also the intervention terms, as you can see in the methods section in the search startegy sub-section.

- **Methods, Search Strategy, p6, line 120: It would be good to know how search strategy was developed, e.g. in conjunction with clinicians/information scientists, adapting a previous search, etc.**

Answer: We specified how the search strategy was developed as follow: "Search strategy was developed by researchers according to the PICOS model and based on the existing literature and finally revised by clinicians."

- **Methods, Search Strategy, p6: Will you also be searching HTA reports for various countries via websites such as NICE? It would also be good to justify why you are restricting your search to PubMed and Embase, when there are other databases. Using only two may make your search less sensitive- how confident are you that all studies will be captured?**

Answer: Dear Reviewer, we understand your point of view. As this study is part of a broader project aimed to investigate on biomarker approach for the optimization of treatment with immune-checkpoint

inhibitors in solid tumors, our first intention is to have an overview of the scientific studies already published in this field. We could consider to investigate also on the HTA reports from various countries in further analysis.

- **Methods, Eligibility criteria, p6, lines 123-130: Check spelling of "Eligibility". As mention under my Abstract comments, it would be good to be clearer about the comparators, and use a PICOS structure, where cost-effectiveness and cost-utility are study designs not outcomes (but with the caveat that adding a study design criteria makes search less sensitive)**

Answer: We've implemented the PICO Model stucture into PICOS as specified above and in the abstract, text and Table 3.

- **Methods, Eligibility criteria, p7, line 132: I couldn't find mention of the date restriction in the search strategy appendix. Perhaps you could add this in. Also do you have plans to update the search if there are delays to the review/prior to publication?**

Answer: Thanks, we have also included the date restriction in the eligibility criteria (between June 2010 to June 2021). In addition, the originally planned date was between June 2010 and June 2020 since we submitted this protocol in December 2020. Given the time between the protocol and the first publication of the review, we have extended the time interval to June 2021.

- **Methods, Eligibility criteria, p7, lines 135-137: These are exclusions by study type, which reinforces that you are using more of a PICOS structure. You might want to consider a summary table of inclusion and exclusion criteria for clarity.**

Answer: Thanks for the advice. We added a specific table (Table 2) which summarize all the selection criteria (inclusion and exclusion).

- **Methods, Eligibility criteria, p7 lines 138-140: if only full texts in English are included, this is a language restriction. It is misleading to begin this sentence saying there is no language restriction**

Answer: Yes, we understand the mistake and rephrased as follow: "Moreover, language restriction will be applied to the research, as fundamental to the eligibility of the study will be the availability of the papers' full text published in English".

- **Methods, Selection and data process, p7, line 146: Please describe how the 4 researchers will divide up screening. Will all 4 review everything, or will there be double screening of sections of the search? What will be your exclusion criteria at the title/abstract stage compared to full text?**

Answer: We specified how we will organize the selection and data process as follow: "Four researchers will double screen titles and abstract to discard irrelevant ones in the first screening phase. Then, full texts of the records selected from their titles and abstracts will be double screened to assess the eligibility for the inclusion in the qualitative analysis. Finally, the references obtained will be validated by clinicians and researchers in the fields of pharmacology, immunotherapy, pharmacovigilance, pharmacoeconomic."

- **Methods, Selection and data process, p7, line 149-150: backward reference searching is described here, but will you also do forward citation searching (seeing what studies cite that paper)? Will there also be searching of grey literature (like the unpublished HTA reports described above)?**

Answer: The backward reference searching will be used as for the forward citation searching. We added this aspect into the text. Anyway, grey literature will be not included in the search as for the unpublished studies or reports. What we would like to obtain in conducting this review is what is already well-known and published in the scientific research about the use of predictive biomarkers in solid tumors treated with ICIs.

- **Methods and Analysis, Selection and data process, p8, lines 154-158: You may also wish to consider the use of an economic specific checklist like CHEERS. The EQUATOR network https://www.equator-network.org/ is good for identifying relevant checklists for different paper types that may help you evaluate published studies as much as write your own**

Answer: Thanks for the suggestion. As abovementioned, the quality of the economic evaluations that will be included in the study will be assessed through the CHEERS checklist, as added in the eligibility criteria sub-section.

- **Methods and Analysis: I would recommend producing for the appendix an example of your data extraction tables and any summary tables that will compare study characteristics, results and study quality. You can then reference these in the methods section**

Answer: The table 3 already reports all variables which will be firstry extracted in Excel. In addition we have added in the Appendix 2 an idea of the summary results table.

- **Ethics and Dissemination: I think this section reads more like part of the methods, with some explanation of how you will present/interpret the results and therefore the heading does not seem appropriate. You may wish to write a separate section focussed on ethics and dissemination, and title this section as part of the methods (evidence synthesis?)**

Answer: We really agree with your observation. We have reviewed this section and incorpored in the Methods section as a sub-section called “Data synthesis”

- **Ethics and Dissemination, p8, line 168-169: better cost-effectiveness will depend entirely on how the study judges cost-effectiveness, e.g., willingness to pay thresholds, what the comparator was. It would be good to describe how you will synthesise different measures of cost-effectiveness particularly across different healthcare systems.**

Answer: We agree that policy in different healthcare systems are heterogeneous. In the table structure in the Appendix 2 we we hypothesized how to synthesize the results and counterbalance the different aspects that emerged from economic evaluations in different health systems.

- **Ethics and Dissemination, p8, line 174: “diffusion of real use” How do you intend to assess this? E.g., will you be comparing trial based evaluations to registry based evaluations to**

evaluations modelled using estimates from published literature and comparing the use to guidelines produced in the country of the study?

Answer: We expect to estimate how widespread is the use of predictive biomarkers in immunotherapy in different countries and to highlight if there is not enough evidence regarding their use and the savings resulting from them both in terms of clinical efficacy and economic savings for health systems

- **Ethics and Dissemination, p9, line 183: This is the first real mention of how you might synthesise the data. Please expand upon this with details of what study characteristics, outcomes you might synthesise and how this might be achieved if the studies are too heterogeneous to warrant a full meta-analysis (could you split into anticipated subgroups for analysis, or would you narratively synthesise everything?)**

Answer: We exploded this concept by adding how we imagine to synthesize the results as follow: "Based on the results obtained, any heterogeneity of the studies will be managed by grouping, if feasible, the included records into different classes such as solid tumor type (e.g., breast cancer, bladder cancer, cervical cancer, colon cancer, head and neck cancer, hodgkin lymphoma, liver cancer, lung cancer, renal cell cancer, skin cancer, stomach cancer, rectal cancer) and study design type (e.g., cost-effectiveness analysis, cost-utility analysis, net-monetary benefit). The same variables present in Appendix 2 will be evaluated for each group and subgroup."

- **References, p11-14: add PRISMA 2020 and refer to it in the methods study development. Also you should address in the text how your study differs from Oosterhoff et al. 2016**

Answer: We provided to add reference to the PRISMA 2020 statement guidelines and also specified differences between our study a that of Oosterhoff et al 2016. Hence, our study differs to that of Oosterhoff et al as they aimed to widely investigate the methodological characteristics of economic evaluations on biomarkers and examine economic aspect. While our systematic review is aimed explore the health-economic impact of predictive biomarkers used in treatment of solid tumors with ICIs comparing them with other targeted and non-targeted therapeutic strategies that do not include the use of the reference biomarker. The final aim is to provide guidance to clinicians and policy makers for the choice of the most effective and sustainable option.

- **Tables, Table 1, p15: It would be good if the search strategy and the PICO criteria lined up, currently the relationship is not clear. You might also want to add the date restrictions, and expand some of the search terms. For example, quality of life might be capture with synonymous terms/measures etc., such as health utility, quality adjusted life years, QALYs, utility, patient reported outcomes, EQ-5D, SF You are also missing several stated outcomes, such as life years (or LYs, life years gained, LYG), net monetary benefit, net health benefit, ICERs. Cost-benefit analysis, cost-effectiveness, cost-utility and economic evaluation are all study types, not outcomes and while it may be justified to include them to make your search more specific, it will make your search less sensitive. I think you should also consider what you are interested in as comparators and include them in your search.**

Answer: Thanks for the advice. As a result of the revision of the entire study protocol, we have made implementations to the search strategy, modifying both Table 1 and Appendix 1.

- **Reporting checklist for protocol of a systematic review: Potential amendments are not explicitly stated in the text**

Answer: We added a specific declaration in the Methods section stating that the PRISMA-P 2015 (Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols) statement checklist was used for the writing of the study protocol. Modifications in the item sequencies were done where appropriate.

- **Objectives: comparators and outcomes are not clear (comparators not specific, study designs incorrectly labelled as outcomes)**

Answer: As abovementioned, we provide to entirely revise the protocol and the PICO Model, implementing it and changing into PICOS to give more importance to the study design of the papers to select. Please, see Table 1 to see the changement done in the model.

- **Data items: no pre-planned data assumptions or simplifications are stated, nor specific variables of interest (e.g. ICER, willingness to pay, costs)**

Answer: We have provided to add a section "Data synthesis" so we've exploded this aspect in order to be more clear and detailed about the hypotetic data synthesis of the results obtained.

- **Outcomes and prioritization: As above, it is not clear which outcomes are to be prioritised. Cost/QALY gained? Net health benefit? Costs?**

Answer: The main results will strictly depend on the study design of the paper (CEA/CUA/BIA ect). So, we would like to highlight first the methodology used in the study (specific model etc) and to report for each biomarker used their cost/QALY gained, willingness to pay (WTP) with the reference threshold as well as the cost-effectiveness profile.

- **Data synthesis 15a-15d: There is very little description of data synthesis in the text, no description of summary measures, or potential subgroups or alternatives to quantitative synthesis if that becomes inappropriate**

Answer: We have improved this by implementing the paper with more information regarding data synthesis process both in the methods section in the "Data synthesis" sub-section and by adding an additional table (Appendix 2).

- **Despite my requests for clarification, I do believe this systematic review could become an informative piece of work. However, at the moment I think more needs to be done in defining the methodology and how the results will be synthesised. I also have concerns that the current search strategies are not suitably justified and may miss some publications. It would be good to look at previous economic search strategies (either in publications or unpublished HTA reports) or involve an information scientist to help identify additional terms for the searches if not done already.**

Answer: From all the authors, we thank you for the helpful advice arising from the review of the entire paper. We definitely revised the entire study protocol by modifying the PICO model and implementing the search strategy in general. The strategy was shared with the researchers and clinical authors of the paper. We have also structured a hypothetical summary of the data analysis more in depth and added a specific section to describe it.